

The Bridge Experiment: Does an Informal $\leftrightarrow$ Formal Join Move End-Task Lean Performance?

Jack McCarthy

WikiLean · <https://wikilean.jackmccarthy.org>

with experiments executed by Claude Code agents under direction

Report, version 2 (**DRAFT**) — July 31, 2026 — post-review corrected edition

DRAFT — pre-adversarial-review edition. Two data slots are open: the blind-judge equivalence tables (§5) and the bare-union ablation (§6.3). Not for citation.

Preregistration: `docs/research/BRIDGE-EXPERIMENT.md`, commit `0d36f2664ab2ebb2c7b80b350d3c9bd820414335` (2026-07-16). Version 1 (2026-07-25) received an external review on 2026-07-27 (`docs/research/review/REVIEW-1.md`), whose claims we verified independently (`verification-of-review-1.json`: 29 confirmed, 1 partial, 1 refuted); this edition incorporates the corrections, and v1 remains in git history. All data, code, and run transcripts live in the private repository `Deicyde/wikilean-bridge-experiment`; every number below is recomputable from a file there (Appendix C).

Registers. [**PREREGISTERED-MODIFIED**] — a preregistered analysis run with a named modification; [**EXPLORATORY**] — not preregistered (all of Bridge v2, the uncalibrated judge); [**CORRECTIVE**] — a reanalysis finding of this edition. **Zero results are cleanly confirmatory:** the preregistered criterion — $D > E$ on *faithful@budget* (semantic equivalence to gold $\wedge$ typecheck) and $D \leq E$ on tokens-to-solve — was never graded on either half (§3); §5 is the exploratory substitute for the missing equivalence leg.

1 Abstract

We test whether a curated dictionary between mathematical concepts and Mathlib declarations (the WikiLean “Brain”) improves language-model performance on formalization tasks, against controls holding the same corpora unjoined. Nothing is cleanly confirmatory — the preregistered semantic-equivalence endpoint was never graded — so our primary metric is the **grounded typecheck rate** (produced declaration, zero hallucinated citations, passing typecheck), which does not measure semantic correctness. On 100 post-Brain-index tasks (held out from the Brain’s indexes but not, we later verified, from the formal-search arms’ sources), the bundled Wikibrain arm reaches 40.6% against 23.2% for the unjoined-tools control on the 69 completed pairs (McNemar 18/6, exact $p = 0.023$, risk difference +17.4pp [+4.1, +30.7]); after repair of the control’s 31 infrastructure-failed runs, the full-100 contrast is 42.0% vs. 30.0% ($p = 0.073$, RD +12.0pp [+0.2, +23.8]). Join-specific attribution awaits a factorial ablation; the clean independent finding is the existence verifier, cutting hallucinated citations to 6.8% against 17.7–21.2% in every other arm. Exploratory: a post-hoc, benchmark-informed union-plus-manual agent took the highest rows we observed on MathlibQR-810 and MathlibMPR (declaration-clustered CIs exclude the single-arm baselines); on SorryDB, kernel-verified proving with and without the Brain is indistinguishable (10 vs. 9 of 171, $p = 1.0$).

2 The system under test: the WikiLean Brain

The Brain is a curated knowledge graph joining Wikipedia/Wikidata concepts, Mathlib4 declarations, and external math databases (nLab, LMFDB, Stacks, ProofWiki, and others). Its atom is a *cell*: a Lean declaration, a Wikidata concept, an article, and a database page denoting the same object are *organs* of one cell, not separate nodes (Mathlib’s `Module`, Wikidata’s “module” and its “vector space” are one atom — Mathlib has no `VectorSpace`). About 12,000 of Mathlib’s declarations are concept-joined this way; the Brain is a map of Mathlib, not the territory. Agents reach it through an MCP server exposing eight tools:

brain_bridge (free text in, existence-verified declarations out — the informal→formal entry point); six graph/content tools (**brain_search**, **brain_cell**, **brain_transfer**, **brain_neighborhood**, **brain_snippets**, **brain_filter**); and **decl_exists**, a batch existence check against the full declaration index — all of Mathlib, not just the joined subset — meant to be called on every name an agent is about to cite. The hypothesis: the informal↔formal *join* is the scarce artifact — a model holds both corpora loosely and hallucinates at their boundary — so curating it should move end-task performance, not just lookup.

3 Preregistered design and what actually ran

Hypothesis. P1: an agent with the *joined* dictionary beats one given informal and formal search *separately*. P2: the bridge solves at lower cost. P3: it lifts grounding, not just compile rate. Success criterion: $D > E$ on faithful@budget ($p < .05$) AND $D \leq E$ on tokens-to-solve.

The five arms. Identical model, prompt, and budgets within each phase; only the tool manifest differs (**bench/run_bridge.py**, **bench/arms/**); see Table 1.

Arm	Tools	Isolates
A no_tools	none (-tools "")	floor / memorization
B informal	Wikipedia + nLab search/fetch	informal reasoning alone
C formal	loogle + ripgrep + source read (Mathlib checkout)	the LeanSearch-class status quo
D wikibrain	the Brain MCP — all eight tools of §2	the join + the existence verifier (not separable here)
E B+C unjoined	B’s tools AND C’s tools, no join	the unjoined control

Table 1: The five preregistered arms.

Two design facts up front. D alone holds **decl_exists**, and hallucination-free citation is a conjunct of the primary metric — D vs. E therefore measures the *bundled Wikibrain package*, not the join (715 **decl_exists** attempts on the 100 fresh tasks, 682 successful). And E failed its manipulation check: 4 informal-tool touches across its 100 fresh runs against B’s 345 — a formal-search agent with a larger manifest.

Models. Tier 1: **claude-haiku-4-5** (model id **claude-haiku-4-5-20251001**), all five arms; Bridge v2: **claude-sonnet-5** (a pre-execution decision, **docs/research/BRIDGE-V2-BENCHMARKS.md**). One model per phase, one seed per (task, arm) — no run-to-run variance is measured anywhere.

Grading. The primary metric is the **grounded typecheck rate**: the run produced a declaration, cited zero names absent from a union oracle (doc-gen4 declaration data $\cup$ verified renames; extractor in **bench/score_bridge.py**), and the declaration typechecked on the pinned toolchain.¹ It does **not** establish that the statement formalizes the prompt — a passing declaration can be weaker, stronger, vacuous, or about the wrong object; the preregistered faithful@budget (BEq+ equivalence to gold) was never graded, its leg in **score_bridge.py** a stub returning **None**.

Task sets. Tier 1a is ProofNet#: 371 problems (341 eval + 30 dev), pinned to **PAug/ProofNetSharp** @a8da405fbd1e348a87445c2e562c747b7e26dc8f (MIT; arXiv:2406.07222), graded on Lean v4.32.0-rc1 / Mathlib a33a5ccd. Tier 1b is the **post-Brain-index fresh set**²: 100 tasks from theorems merged into Mathlib master 2026-07-03→07-16, verified absent from the Brain’s declaration universe and node set — the indexes arm D serves (**bench/data/fresh_tasks.stats.json**) — and graded on Lean v4.33.0-rc1 / Mathlib 9944fe29. It was **not** verified absent from the formal-search arms’ sources: C/E read a mutable checkout (61a5e4f338, content through ~2026-07-10) and live unsnapshotted Loogle (exposure measured in §4.4). The preregistered determinacy pre-screen became a post-hoc double annotation (agreement 79%, $\kappa \approx 0.20$; a 74-task both-determinate subset, none excluded).

¹Version 1 called this metric “success (folded)”; renamed after the review — a metric with no equivalence leg should not be called success.

²Version 1’s “contamination-proof” and “newer than every arm’s index” are both retracted (§9, concern 4).

Execution. A 10-minute wall clock was enforced; the 30-turn budget was **advisory only** (stated in the prompt, no CLI cap) — fresh-set overruns C 50 runs, D 38, E 32, maxima 80/88/72. The fresh arms ran in strictly sequential blocks on 2026-07-19 (A 01:50–02:08Z, B 02:09–03:56Z, C 03:58–04:27Z, D 04:29–04:54Z, E 04:56–05:17Z), so arm is confounded with time. Agents had no in-loop typechecker and were told to verify cited names through their tools.

The arm-E 429 incident and repair [CORRECTIVE]. Rows fresh_069–099 — 31 contiguous tasks at the tail of E’s sequential block — errored with session-limit 429s: infrastructure, not agent behavior. On 2026-07-27 they were rerun (`bench/analysis/rerun_E_fresh.py`) with E’s exact July-19 code path on a read-only git archive of 61a5e4f338, the same tree the original C/E runs saw. The repair surfaced a second hazard: under concurrent cold starts the CLI’s stdio MCP servers can silently fail to connect, yielding runs that *look* like completions but had zero tools; the repair driver detects this from the stream-json init event and retries (commit 19a90209). Originals: `bench/data/runs_E_fresh_429_archive/`; 14 of the 31 repaired rows became grounded-typecheck passes (`bridge_summary_v2.json`, `v2_provenance`).

Execution summary. The components whose status changes the interpretation; the complete 25-component inventory is Appendix B.

Preregistered component	Status → consequence
faithful@budget (BEq+ equivalence)	not executed (grader a stub) → primary endpoint missing; zero confirmatory results
LLM judge + 50-item human calibration tokens-to-solve (half the criterion)	not executed in campaign → uncalibrated post-review pass only (§5)
3 reseeds + pass@k curves	not executed → P2 untested as specified
Second model class on the primary set	not executed → single seed; no variance estimate anywhere
Tier 2 as specified	not executed → Haiku Tier-1 / Sonnet v2; generality unknown
30-turn budget	replaced by exploratory v2 → §§6–7 exploratory
Fresh-set holdout	modified (advisory only) → uncontrolled; §4.3
Sequential blocks; arm-E 429 outage	Brain indexes only → renamed post-Brain-index; §4.4
	corrective repair → snapshot E row conflates outage with behavior; §4.2

4 Tier 1: statement autoformalization (claude-haiku-4-5)

[PREREGISTERED-MODIFIED] — the preregistered paired design, graded on grounded typecheck rate instead of faithful@budget; one model, one seed, advisory budget. Sources: `bench/analysis/tier1_reanalysis.{json,md}`, `part1_fresh100_v2.{json→bridge_summary_v2.json,md}`. All rates carry Wilson 95% CIs; McNemar counts are discordant pairs (tasks exactly one arm passed), written first-arm-only/second-arm-only.

4.1 Tier 1a — ProofNet# eval split (n=341/arm)

arm	grounded typecheck	Wilson 95% CI
A none	204/341 = 59.8%	[54.5, 64.9]
B wiki	198/341 = 58.1%	[52.8, 63.2]
C formal	218/341 = 63.9%	[58.7, 68.9]
D wikibrain	219/341 = 64.2%	[59.0, 69.1]
E B+C unjoined	208/341 = 61.0%	[55.7, 66.0]

The arms sit within a few points of each other (v1’s D-vs-E McNemar, 63 vs. 51 discordant over all 371 pairs, $p = 0.30$, is an underpowered null). Arm A at ~60% with zero tools is substantial ProofNet memorization — the motivation for Tier 1b. Typecheck alone *anti-correlates* with grounding (E led typecheck at 83.3% while trailing on hallucinations), reproducing TheoremGraph’s typecheck-is-not-a-signal finding at $15\times$ their n . Contaminated by construction, Tier-1a serves only for paired deltas and the memorization control.

4.2 Tier 1b — the post-Brain-index fresh set (n=100/arm)

Three analysis bases, one table: E’s 31 infrastructure-dead rows counted as failures; the 69 pairs arm E completed; the full 100 after the repair (§3). Cells are grounded typecheck, $n/N = \%$ [Wilson 95% CI].

arm	errors-as-failures (n=100)	completed pairs (n=69)	post-repair (n=100)
A	20/100 = 20.0% [13.3, 28.9]	10/69 = 14.5% [8.1, 24.7]	20/100 = 20.0% [13.3, 28.9]
B	22/100 = 22.0% [15.0, 31.1]	12/69 = 17.4% [10.2, 28.0]	22/100 = 22.0% [15.0, 31.1]
C	25/100 = 25.0% [17.5, 34.3]	12/69 = 17.4% [10.2, 28.0]	25/100 = 25.0% [17.5, 34.3]
D	42/100 = 42.0% [32.8, 51.8]	28/69 = 40.6% [29.8, 52.4]	42/100 = 42.0% [32.8, 51.8]
E	16/100 = 16.0% [10.1, 24.4] — 31 in-fra errors	16/69 = 23.2% [14.8, 34.4]	30/100 = 30.0% [21.9, 39.6]

D vs. C (28/11 discordant, exact $p = 0.0095$) and D vs. A (29/7, $p = 0.0003$) are untouched by E’s outage. D vs. E: on completed pairs — the headline Tier-1 contrast — 18/6, $p = 0.023$, RD +17.4pp [+4.1, +30.7] (paired Wald; D vs. C 21/5 $p = 0.0025$ and D vs. A 22/4 $p = 0.0005$ there); post-repair full-100, 25/13, $p = 0.073$, RD +12.0pp [+0.2, +23.8]. The spread: significant on completed pairs, marginal on the full set (the RD interval barely excludes zero; the McNemar p misses 0.05); the errors-as-failures column conflates an outage with behavior, and its inflated p survives only in §9. Repaired, E is mid-pack (30%, indistinguishable from C, $p = 0.42$); v1’s “unjoined tool volume can be actively harmful” is retired. Arm A’s collapse from 59.8% (1a) to 20% is the study’s cleanest memorization quantification.

4.3 Turn-budget sensitivity [CORRECTIVE]

Restricted to pairs where both arms stayed within 30 turns (tier1_reanalysis.md §5): D vs. E $n = 45$, 46.7% [32.9, 60.9] vs. 24.4% [14.2, 38.7], 16 discordant, $p = 0.021$ — but that counts E’s 429 rows (turns=1) as within-budget failures, and the completed-only version is underpowered ($n = 27$, 44.4% [27.6, 62.7] vs. 40.7% [24.5, 59.3], 7 discordant, $p = 1.0$). D vs. C within budget: $n = 35$, 54.3% [38.2, 69.5] vs. 25.7% [14.2, 42.1], 12 discordant, $p = 0.0063$. Consistent in direction; for D-vs-E, inconclusive once both restrictions apply at once.

4.4 Exposure strata: what the formal-search leak touched [CORRECTIVE]

We measured the §3 leak against the pinned tree 61a5e4f338 (bench/analysis/fresh_exposure.{json,md}): of the 100 golds, the full dotted name is in-tree for **37**, the basename is a declaration header anywhere for **64** and in the task’s own module file for **51** (the stratum basis), and the gold’s commit is an ancestor of the pin for 49.³ Cells: grounded typecheck, snapshot basis — E’s 429 rows count as failures, hitting its unexposed cell hardest; attempted-only excludes them.

arm	exposed ($n = 51$)	unexposed ($n = 49$)
A	9/51 = 17.6% [9.6, 30.2]	11/49 = 22.4% [13.0, 35.9]
B	8/51 = 15.7% [8.2, 28.0]	14/49 = 28.6% [17.8, 42.4]
C	11/51 = 21.6% [12.5, 34.6]	14/49 = 28.6% [17.8, 42.4]
D	24/51 = 47.1% [34.1, 60.5]	18/49 = 36.7% [24.7, 50.7]
E	12/51 = 23.5% [14.0, 36.8]	4/49 = 8.2% [3.2, 19.2]
E attempted-only	12/44 = 27.3% [16.4, 41.9]	4/25 = 16.0% [6.4, 34.6]

McNemar by stratum: D vs. E exposed 15/3 $p = 0.0075$, unexposed 17/3 $p = 0.0026$; D vs. C exposed 18/5 $p = 0.0106$, unexposed 10/6 $p = 0.45$. The leak’s direction favors C/E, and D’s edge over E is strongest exactly where there was nothing to leak (36.7% vs. 8.2% unexposed; attempted-only, vs. 16.0%) — so the leak cannot explain that contrast. D over C, by contrast, concentrates in the exposed stratum and is a null unexposed — part of D’s advantage over pure formal search may rest on targets C could have found in-tree but didn’t.

4.5 Hallucinated citations and trace attribution

Fresh set, all 100 rows per arm, post-repair (union oracle of score_bridge.py):

arm	halluc-decl rate (cited names)	runs w/ ≥ 1 halluc.	Wilson 95% CI (runs, $n = 100$)
A	94/443 = 21.2%	54	[44.3, 63.4]
B	76/429 = 17.7%	48	[38.5, 57.7]
C	108/517 = 20.9%	49	[39.4, 58.7]
D	32/472 = 6.8%	23	[15.8, 32.2]
E	107/505 = 21.2%	49	[39.4, 58.7]

³Measured on tree bytes, not merge metadata: **fresh_037** and **fresh_054** are exposed despite post-pin commits (**fresh_054**’s dotted name sits in its module file at the pin) — hence 51 exposed $\neq$ 49 merged-before-pin.

Citation-level rates are descriptive (citations cluster within runs); the run-level proportions carry the CIs. This is the study’s most robust effect: unchanged by the repair, and larger off-distribution than on (Tier-1a: D 5.9% vs. 10.1–11.3%). **Trace attribution [EXPLORATORY — eval split only]** (`bench/trace_analysis.py`): 98–100% of traced tool-arm runs cite at least one declaration that surfaced in a tool result; only 10–17% of citations never touched the tools; 35% of arm-D eval runs (116/329) cite a declaration from a `brain_bridge/brain_transfer` result — English in, formal name out (an undercount: result heads truncate at 200 chars). The sharpest signal is among *hallucinated* citations: the checked-and-cited-anyway rate is 35% for C and 30% for E but **13% for D** — binary verification disciplines the model where fuzzy search lets it fool itself.

4.6 The existence verifier as an independent finding

What the data support: a **batch, binary, oracle-backed existence check**, used on every name, collapses hallucinated citations roughly $3\times$ across two task distributions, concentrated on citations the agent had already “checked” by search. What they do not yet support: how much of D’s grounded-typecheck edge is the verifier versus the join — D alone holds it, and no-hallucination is a conjunct of the metric; the 2×2 factorial (§10) separates them. One circularity: `decl_exists` and the scoring oracle draw on the same doc-gen4 index, so “fewer hallucinations” partly means “the arm could query the grader’s oracle” — itself the product claim, but it must be named.

Cost (descriptive only). Campaign means per task over all 471 runs/arm as run (pre-repair; E’s 31 error rows cost ≈ 0): A \$0.034, B \$0.048, C \$0.140, D \$0.121, E \$0.128; mean wall-clock C 116 s, D 89 s, E 106 s. tokens-to-solve was never computed (§3), so P2 is untested; v1’s “cheaper and better” framing is withdrawn.

5 Semantic faithfulness (uncalibrated judge) [EXPLORATORY]

Following the review, a blind LLM-judge pass over all 500 fresh outputs (5 arms $\times$ 100 tasks, the 31 repaired E rows included) was launched 2026-07-27 (`bench/analysis/judge_fresh_run.py`) as the exploratory substitute for the ungraded equivalence endpoint (§3). Judge = `claude-sonnet-5`, deliberately stronger than the Haiku subjects; no tools, one turn; prompt = {informal statement, gold, produced declaration}, scanned for arm-revealing substrings (abort on hit); no-output rows count not-equivalent; a fixed 50-item seed-stratified subset (seed 20260727, 10 per arm) is re-graded for self-consistency. Pre-declared deviation: the judge sees `gold_context` + `gold_formal` (the `variable/open` binders), arm-independent. **The judge is uncalibrated** — the preregistered 50-item human calibration remains undone — so nothing here rescues the missing confirmatory endpoint.

Pending: blind-judge equivalence tables (grading in progress)

[[SLOT:JUDGE]] — to be filled from `bench/analysis/judge_fresh_summary.json` when the pass completes (arms A–D graded, E in progress). Tables to land, per `judge_fresh_summary.py`: (1) per-arm strict and evaluated equivalence rates on fresh-100, Wilson 95% CIs; (2) the conjunction grounded typecheck $\wedge$ judge-evaluated-equivalent — the closest available analogue of faithful@budget; (3) exact-binomial McNemar D-vs-E, D-vs-C, D-vs-A on both, full-100 and completed-69; (4) self-consistency on the 50-item re-grade, plus the judge cost/error census.

6 Third-party retrieval: MathlibQR-810 and MathlibMPR (claude-sonnet-5) [EXPLORATORY]

Bridge v2 uses graders that predate the arms: third-party expert gold labels here, the Lean kernel in §7. Data: `frenzymath/LeanSearch-v2 @94f4888cbaf9` (CC BY 4.0; arXiv:2605.13137). Arms: N (no tools), F (loogle + `decl_grep` + `decl_read`), W (the Brain MCP), and WF ($W \cup F$ **plus** the Appendix-A manual prepended to the prompt). Scoring: `bench/v2/score_retrieval.py`, exact full-name match.

WF is a post-hoc, benchmark-informed condition; every WF number carries that label. Verified commit times: N/F/W results 22:27 on 07-24; the manual 00:30 on 07-25; WF results 01:15. The manual distills measurements from the same evaluation queries WF was then scored on — its bracketed

figures (per-style nDCG, the 63% brain_cell failure rate, the 143/810 format failures, call counts) are the N/F/W/system-mode eval-set aggregates, reproduced exactly — so WF is development evidence of a maximally equipped, briefed agent, not an untouched test result or a defensible SOTA comparison. (Version 1 also misattributed the 63% figure to “the Tier-1 traces”; it is the v2 W-arm figure — 482 calls, 63.3% failed.)

6.1 MathlibQR fair-810 — concept retrieval, declaration-clustered [CORRECTIVE statistics]

The 810 query rows are paraphrase styles of only **171 distinct gold declarations** (2–6 rows each), so inference is at the declaration level (`bench/analysis/retrieval_clustered.md`; cluster bootstrap $B = 10,000$, seed 20260727):

system	R@10 (row)	R@10 95% CI (decl-cluster boot)	nDCG@10	nDCG 95% CI
published: TheoremGraph	0.775	—	0.548	—
published: LSv2 retriever+reranker	0.780	—	0.623	—
system-mode wikibrain (one brain_bridge call, no LLM)	0.036	—	0.031	—
agent N	0.633	[0.581, 0.684]	0.598	[0.547, 0.647]
agent F	0.831	[0.792, 0.868]	0.790	[0.750, 0.829]
agent W	0.816	[0.767, 0.862]	0.781	[0.731, 0.827]
agent WF (post-hoc, benchmark- informed)	0.885	[0.849, 0.919]	0.839	[0.801, 0.874]

Declaration-level paired contrasts: **F – W is a null** (R@10 +0.015 [−0.028, +0.059], Wilcoxon $p = 0.26$; nDCG +0.010 [−0.029, +0.048]) — tied, with different textures: W wins the special_case style (nDCG 0.523 vs. 0.384), F the Lean-syntax styles (grep cannot find what source never names). **WF – F +0.054** [+0.027, +0.082], $p = 0.0017$; **WF – W +0.069** [+0.032, +0.108], $p = 0.0003$ — the briefed union beats both single arms even under clustering.⁴ The system-mode row is the headline API deficiency: one brain_bridge call scores 0.036 — the free-text entry is a label/alias resolver, not a semantic retriever — while an agent *iterating* over the same API reaches 0.816; the content is in the graph, the single-shot entry point is the gap. N’s 0.633 includes 143/810 rows scored 0 for format non-compliance.

6.2 MathlibMPR — premise retrieval (69 post-cutoff PR theorems)

One task per PR, so no cluster correction is needed (§9); task-level bootstrap CIs:

system	group-recall@10	95% CI (task boot)
published: LeanSearch-v2 reasoning	0.461	—
published: DIVER	0.380	—
published: TheoremGraph concept-tuned	0.165	—
system-mode wikibrain	0.000	—
agent N	0.203	[0.131, 0.282]
agent W	0.272	[0.196, 0.354]
agent F	0.453	[0.365, 0.540]
agent WF (post-hoc, benchmark-informed)	0.557	[0.472, 0.642]

A generic Sonnet agent with grep and loogle **matches the specialist premise retriever** (0.453 vs. 0.461), untouched by this study’s confounds. W trails F by 18pp: the preregistered concept $\neq$ premise boundary (TheoremGraph’s negative transfer) measured on our own tools — a quantified target for **brain_premises** (BRIDGE-ISSUES #7). **WF – F is marginal**: +0.104 [+0.007, +0.201], sign test $p = 0.029$, Wilcoxon $p = 0.049$.

6.3 Tool use under the manual

Census, pooled QR+MPR: F averages 2.2 calls/run; W 3.5 (decl_exists 1,251 + brain_bridge 608 — verify-then-cite made mechanical); WF 4.6 in a genuine dual-toolkit mix (decl_grep $\sim$ 1.5k, decl_exists $\sim$ 0.9k, brain_bridge $\sim$ 0.7k, decl_read $\sim$ 0.5k, loogle $\sim$ 0.4k). Under the manual, brain_cell misuse (63% failure in W) falls to a single call in WF. Same-eval-set caveat as above.

⁴Version 1’s row-level McNemars ($n = 810$ treated as independent: F–W 78/66 discordant $p = 0.36$; WF–F 71/27 $p = 1.0 \times 10^{-5}$; WF–W 83/27 $p = 8.3 \times 10^{-8}$) are uncorrected for the clustering — the 144 F/W discordant rows come from only 76 declarations — and are superseded.

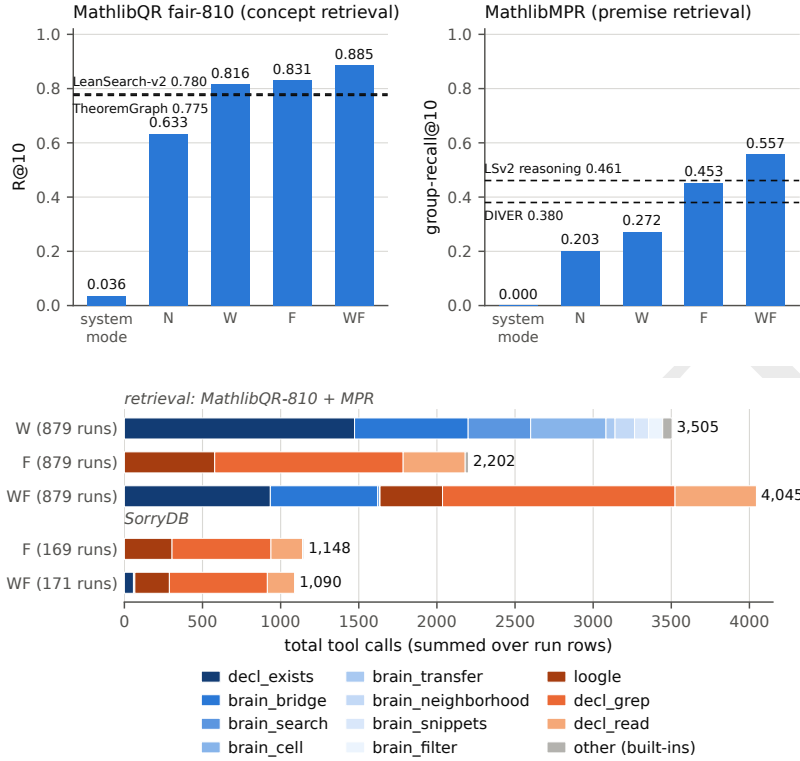


Figure 1: Top: retrieval point estimates (published anchors dashed); left QR fair-810 R@10, right MPR group-recall@10. WF is the **post-hoc, benchmark-informed** condition (§6); point estimates unchanged from v1, and the redone declaration-level CIs are in the tables, not drawn here. Bottom: tool composition by arm (calls summed over run rows; retrieval pools QR-810 + MPR, 879 runs/arm; SorryDB ~170). Blues = Brain tools, oranges = formal search, gray = CLI built-ins. Under the manual (WF) the Brain toolkit contracts to **brain_bridge** + **decl_exists**; on SorryDB WF routes 94% of calls to formal search (§7). Telemetry unchanged by this edition’s corrections.

Pending: bare-union (U) ablation (queued)

*[[SLOT:UNION]] — the bare-union U arm ($W \cup F$ tools, no manual; **MANUAL_ARMS** blanked in `bench/v2/run_agent.py`) separates the union effect from the briefing effect: N/F/W/U/WF on both benchmarks, declaration-clustered (QR) and task-bootstrap (MPR) CIs, U-vs-F and WF-vs-U paired contrasts. Until it lands, “union tools” and “benchmark-informed manual” remain unseparated in WF.*

7 SorryDB: kernel-graded proving in live repositories (claude-sonnet-5) [EXPLORATORY]

SorryDB (arXiv:2603.02668, <https://sorrydb.org>) serves unsolved `sorry`s from real Lean repositories; success is the repo *building* with your proof — no judge, nothing to memorize. From the SorryDB-2601 1,000-task evaluation split we sampled repositories (disk allowed one repo build at a time). **Snapshot rot:** the original freeze (8 repositories, 164 tasks) lost 74 of 164 tasks (45%) in ~six months — pinned commits of GlimpseOfLean, LeanCourse25, and most Foundation and SemicircleLaw tasks no longer exist on GitHub (history rewritten) — and the rot is invisible without actually fetching every pin: GitHub’s archive endpoint answers preflight HEAD for dead commits with HTTP 200. The refreeze (`bench/v2/sorrydb_prep.py`, commit `db679e8d`) keeps verified-fetchable pins only: **171 tasks across 10 repositories**. Protocol: build-free, one-shot — goal state plus a ± 60 -line window at the pin; the agent (600 s; arms N/F/WF as in §6) outputs one replacement for the `sorry` under an explicit honest-abstention rule; `bench/v2/verify_sorrydb.py` clones, splices, rebuilds; success = exit 0 with no `sorry/axiom` warning.

Results — verdict-complete (all 203 frozen-task candidates have kernel verdicts, commit 9682b3b1; v1’s 8 curve25519 stragglers verified as 0 proved); $n = 171$ tasks/arm:

	N no tools	F formal	WF union+manual (post-hoc)
run rows present	168	169	171
no output (timeout / no lean block)	70	15	11
honest give-up (literal <code>sorry</code>)	40	83	86
candidate proofs submitted	58	71	74
proved (kernel)	2	9	10
failed	50	55	57
unspliceable	5	7	7
verify timeout	1	0	0
proved / 171	1.2%	5.3%	5.8%
Wilson 95% CI	[0.3, 4.2]	[2.8, 9.7]	[3.2, 10.4]
repo-clustered boot 95% CI	[0.0, 2.7]	[0.0, 10.7]	[0.0, 12.2]
proved / candidates	3.4%	12.7%	13.5%
total cost (USD)	\$58.97	\$102.76	\$108.87
cost per proved	\$29.48	\$11.42	\$10.89
mean wall-clock / task	120 s	198 s	183 s

(3 N and 2 F tasks lack run rows — runner losses. Distinct sorries proved: 13; N’s are a strict subset of F’s and WF’s; $WF \cap F$ 6, WF-only 4, F-only 3; proofs concentrate in PersistentDecomp, LeanAPAP, MiscYD — 3 of the 10 repos.)

WF vs. F is statistically indistinguishable: one proof apart (10 vs. 9 of 171), task-level exact McNemar 4/3, $p = 1.0$; repo-clustered bootstrap +0.58pp, 95% CI [+0.0, +1.85pp], 34% of resamples no difference or worse (`retrieval_clustered.md` §4). SorryDB says essentially nothing about the Brain — WF routed 94% of its tool calls to formal search, and no W-only or bare-union arm ran. It does show the tools-vs-no-tools gap: N burns its budget (70 of 168 rows produce nothing, mostly 600s timeouts) while tool arms triple-to-quintuple the prove rate and double honest abstention — though 10 repo clusters warrant caution even there. Cost-per-proved is bookkeeping only (unstable at 2/9/10 successes); v1’s “adding the join never made the task more expensive per success” is deleted (§9). The published SorryDB agentic best ($\sim 30\%$) is not comparable: different task subset, different protocol (published agents iterate against the build; ours drafted one-shot).

8 What survives and what does not

Each retired v1 claim, with its v2 replacement:

v1 claim (retired)	v2 replacement (supported)
“The join carries the effect” (fresh McNemar $p < 0.0001$) “42% success”	The bundled Wikibrain condition outperforms the tested controls (§4.2); join attribution awaits the 2×2 factorial “42% grounded typecheck rate ” — no equivalence leg graded; §5 is the exploratory substitute
“Contamination-proof / newer than every index”	“ Post-Brain-index ”: Brain indexes only; 51/100 golds exposed to C/E, outcomes stratified (§4.4)
“Arm E worst at 16%; unjoined tool volume actively harmful” “WF sets the best rows we know of”	The 31 no-output rows were 429s; repaired, E is mid-pack at 30% (§4.2) A post-hoc, benchmark-informed condition took the highest observed rows (QR 0.885, MPR 0.557); manual tuned on the same eval queries (§6)
“Adding the join never made the task more expensive per success” brain_cell 63% failure “in the Tier-1 traces”	Deleted — cost is descriptive, tokens-to-solve never computed, SorryDB 10-vs-9 $p = 1.0$ The figure is from the v2 W-arm eval runs (§6)

What survives, with register: the **hallucination collapse** / existence-verifier finding (§4.5–4.6) [preregistered-modified + exploratory traces]; the **memorization quantification** (59.8% $\rightarrow$ 20.0%) [preregistered-modified]; the **system-mode gap** (0.036 vs. 0.816, §6.1) — an API deficiency, not a content deficiency [exploratory]; the **F-matches-SOTA baseline** and **concept $\neq$ premise boundary** (§6.2) [exploratory]; **snapshot rot** (§7) [exploratory]; and the **tools-vs-no-tools proving gap** (§7) [exploratory, 10 clusters].

9 Response to external review 1

The review raised eight concerns; all eight were confirmed in substance after claim-by-claim verification, and the table gives the action on each (full record: [docs/research/review/](#)). Two of its factual claims did not survive (an error rate of two across 31 checked): MathlibMPR does not cluster by PR (69 tasks, 69 distinct PRs — no correction needed; the refuted claim), and arm E’s 31 declaration-less runs were contiguous 429 errors, not interface overload. The second correction cuts both ways: it removes the review’s overload reading, and it removes v1’s error-inflated headline $p = 2.4 \times 10^{-5}$ with the “unjoined volume is harmful” claim (repaired contrasts: §4.2).

#	Concern	Action in v2
1	“Success” measures no semantic correctness	Renamed grounded typecheck rate ; judge pass §5; calibration roadmap #4
2	D vs. E does not isolate the join	“Join carries the effect” retired; bundled-condition language (§4.6); 2×2 factorial roadmap #1
3	Arm E failed its manipulation check	Disclosed (§3); yoked control roadmap #2
4	Fresh set not isolated from formal-search sources	Renamed post-Brain-index ; exposure strata (§4.4); frozen snapshots roadmap #7
5	WF is test-set tuned	Labeled post-hoc, benchmark-informed everywhere; provenance in §6; 63% misattribution fixed; bare-union slotted; manual freeze roadmap #3
6	Independence and execution (clustered rows, sequential blocks, advisory budget, single seed, no CIs)	Declaration-clustered reanalysis (§6.1); Wilson/bootstrap CIs and RD+CI throughout; §3 disclosures; row-level tests demoted to footnotes
7	Preregistered experiment incomplete; deviations list silent	Execution summary + Appendix B inventory; register labels; zero-confirmatory up front
8	SorryDB evidence about the join is thin	All 203 verdicts completed; WF-vs-F indistinguishable; repo-clustered CIs; cost claim deleted (§7)

10 Roadmap

Adopted from the review’s priority list, in its order:

1. **2×2 factorial** — {join, no join} $\times$ {decl_exists, none} on a truly frozen post-cutoff set; turns §4.6’s attribution question into a measurement.
2. **Yoked unjoined control** — one integrated interface, informal and formal panels without declared correspondences, plus a preregistered manipulation criterion.
3. **Manual freeze on development data** — split MathlibQR by its 171 target declarations, evaluate on unseen targets; run bare-union, F+manual, W+manual, union+manual.
4. **Human judge calibration** — the preregistered 50-item hand-graded set before any faithfulness number leaves the exploratory register.
5. **In-loop compiler proving** — the SorryDB follow-up with identical build access per arm.
6. **Multi-seed, multi-model** — ≥ 3 seeds per task and a second model family.
7. **Frozen search snapshots** — pinned checkout predating every test theorem, snapshotted Loogle index, per-run hashes of tree, index, Brain snapshot, prompts, and manifests.

11 Data availability

Everything needed to recompute this report lives in the private preservation repository **Deicyde/wikilean-bridge-experiment** (report at the root as `README.md` and under `report/`); provenance commits refer to the WikiLean working repository. The complete file map, recomputation entry points, and external sources are Appendix C.

A The agent manual

bench/v2/AGENT_MANUAL.md, prepended verbatim to every WF-arm prompt; a **post-hoc, benchmark-informed** artifact (§6) — its bracketed figures were measured on the same MathlibQR/MPR evaluation queries the WF arm was then scored on.

Tool manual: searching Mathlib with the Brain + formal search

You have two complementary toolkits. Every claim in this manual is measured from ~5,500 logged tool calls across 3,500 benchmark runs (Tier-1 + Bridge v2); numbers in brackets are those measurements.

A.1 The mental model

- **The Brain (wikibrain, 8 tools)** is a *curated knowledge graph* joining informal mathematics (Wikipedia/Wikidata concepts, external databases) to Mathlib declarations. It knows what things *mean* and how concepts relate — but it only “atomizes” declarations that some concept, annotation, or tag claims (~12k of Mathlib’s declarations). It is a map, not the territory.
- **Formal search (3 tools)** operates on *all* of Mathlib directly: type-pattern search (loogle), source grep, source reading. It knows everything that exists but nothing about what it means informally.

Rule of thumb: **route by what you’re holding**. Holding a concept described in words → start with the Brain. Holding a type shape or name fragment → start with formal search. Always finish with `decl_exists` on every name you intend to output.

A.2 Formal search tools

A.2.1 loogle

Type-pattern and constant search (the public `loogle.lean-lang.org` service).

- USE for: “a lemma whose statement looks like `_ * _ ≤ _ * _`”, searches by involved constants (`Real.sqrt`, `tsum`), name substrings.
- Strengths: covers ALL of Mathlib; ~0% call errors [578 calls, 0 errors]; the best premise-finding tool — the loogle+grep toolkit matched a specialist premise retriever [group-recall@10 0.453 vs 0.461 SOTA].
- Weaknesses: needs Lean syntax fluency; a wrong pattern silently returns unrelated hits; small result payloads mean you often need `decl_read` next.

A.2.2 decl_grep

Regex search over the Mathlib source checkout (ripgrep).

- USE for: name fragments (“cantor”), notation, docstring phrases, finding the file a declaration lives in.
- Strengths: the workhorse [1,206 calls, 0% errors]; catches things loogle’s type index can’t (comments, docstrings, naming conventions).
- Weaknesses: lexical only — if the concept’s name doesn’t appear in source, grep cannot find it (descriptive queries like “special case of X” score worst for grep-based agents [nDCG 0.384 vs the Brain’s 0.523]).

A.2.3 decl_read

Read declaration source (with surrounding context) from the checkout.

- USE for: verifying a candidate actually says what you think BEFORE citing it; getting exact signatures and hypotheses.
- Strengths: ground truth for any decl in Mathlib [397 calls, 0% errors].
- Weaknesses: you need the file/name first — it is a confirmation tool, not a discovery tool.

A.3 Brain tools

A.3.1 brain_bridge — THE ENTRY POINT for informal→formal

Free text in, existence-verified declarations + owning atoms out, with signatures, import lines, and one-hop dependency context.

- USE for: “the concept described as ⟨words⟩” — especially nicknames and special-case descriptions, where the Brain’s curated bonds beat grep [special_case nDCG 0.523 vs 0.384; nickname 0.779 vs 0.757].
- CRITICAL LIMITATION (measured): its free-text matching is label/alias anchored, NOT semantic. A single call with a descriptive paraphrase usually misses [single-shot benchmark: 3.6% R@10 vs 82% for an agent that reformulates]. **Never accept one miss as the answer: reformulate 2–4 times** — try the concept’s canonical name, a synonym, the head noun alone. 8.9% of calls return `match: “none”` [727 calls]; that response includes nearest-candidate suggestions — read them.
- The `hits` list is existence-verified; bond quality matters: **exact** means the decl IS the concept; **formalizes**/weaker bonds are nearby.

A.3.2 brain_search

Fuzzy label + alias search returning atoms (cells).

- USE for: resolving a name you half-know into an atom id; browsing what the Brain calls something [400 calls, 0% errors — the reliable fallback when `brain_bridge` misses].
- Weaknesses: returns atoms, not ranked decls — follow with `brain_cell` on the winning atom.

A.3.3 brain_cell

The full atom card: Lean code, Wikidata description, DB snippets, organs.

- USE for: everything known about ONE object after bridge/search resolved it.
- **THE #1 MISUSE [63% of 482 calls failed]: calling it with a bare `decl:Mathlib:Name` for an arbitrary declaration.** The Brain only has atoms for concept-joined decls; “unresolvable key” means *no atom owns this decl*, NOT that the decl doesn’t exist. For arbitrary decls use `decl_exists` (works for all of Mathlib) and `decl_read` for source. Call `brain_cell` only with ids that came OUT of a Brain tool.

A.3.4 decl_exists — THE DISCIPLINE TOOL

Batch existence check for declaration names (backed by the full oracle — covers ALL of Mathlib, unlike `brain_cell`).

- **USE ALWAYS, on every name you are about to output.** This is the measured difference between grounded and hallucinated citations: agents that verified before citing cut hallucinated-but-cited-anyway to 13% vs 33% for agents relying on fuzzy search results alone. It is cheap [1,473 calls, 0.1% errors] and batched — check 10 names in one call.
- Also returns verified-rename suggestions for stale names.

A.3.5 brain_neighborhood

The synapse graph around an atom (depends/links/relates, cursored).

- USE for: “what connects to X” — finding the connecting lemma between two concepts, walking dependencies.
- Weaknesses: needs a valid atom id [24.8% of calls errored — same unresolvable-key trap as `brain_cell`; resolve the atom first].

A.3.6 `brain_transfer` / `brain_snippets` / `brain_filter`

Narrower tools: label $\leftrightarrow$ decl jumps for KNOWN labels (transfer), stored source snippets (snippets), facet enumeration (filter). High misuse rates when fed non-atom ids [40%/57% errors]. Prefer `brain_bridge`/`brain_search` entry; reach for these only on ids the Brain already returned.

A.4 Routing playbook

You are holding	Do this
A concept in words	<code>brain_bridge</code> $\rightarrow$ (miss?) re-formulate $\rightarrow$ <code>brain_search</code> $\rightarrow$ <code>brain_cell</code>
A special case / nick-name	<code>brain_bridge</code> (the Brain’s best terrain)
A type shape	<code>loogle</code>
A name fragment	<code>decl_grep</code>
Premises for a proof	<code>loogle</code> + <code>decl_grep</code> FIRST [0.453 vs 0.272], Brain for concept grounding
A candidate name to cite	<code>decl_exists</code> (batch, always) $\rightarrow$ <code>decl_read</code> if the statement matters
An atom id from any Brain tool	<code>brain_cell</code> / <code>brain_neighborhood</code> freely
A bare decl name needing info	<code>decl_exists</code> + <code>decl_read</code> — NOT <code>brain_cell</code>

A.5 Anti-patterns (each one measured in prior runs)

1. **Citing unverified names.** 33% of hallucinated citations came from agents that had search evidence in front of them and cited a wrong name anyway. `decl_exists` is binary and kills this.
2. **One query, then giving up.** The single-call miss rate on descriptive queries is $\sim 96\%$; agents that reformulate reach $\sim 82\%$. Iteration IS the algorithm.
3. **`brain_cell` as a generic decl inspector.** 63% failure rate. Brain atoms $\neq$ all of Mathlib.
4. **Answering outside the requested format.** 143/810 no-tool runs were scored 0 for exactly this. Re-read the output contract before replying.
5. **Inventing tool names** (e.g. `logue`, `lookie`). Use the names above.

B Complete execution inventory

Every preregistered component, its status, and the inferential consequence. (Sources: the preregistration; the claim-by-claim inventory in `verification-of-review-1.json`, prereg section.)

Preregistered component	Status	What actually happened	Inferential consequence
Five arms A–E, identical model/prompt/budget, tools-only difference	executed	471 rows/arm (<code>bench/data/runs/</code>)	design intact
Per-tool call logging	executed	<code>tool_calls_by_name</code> every row; full traces eval C/D/E + fresh (deviation 7)	enables §4.5 attribution
Tier 1a ProofNet# (371)	executed	371/arm, scored	contaminated by construction (arm A 59.8% on eval-341); paired deltas only
Tier 1b fresh set (~ 100)	executed	100 tasks	holdout guarantee narrower than claimed (Brain indexes only) — renamed post-Brain-index
Hallucinated-decl rate	executed	all tables	the cleanest surviving mechanism signal
McNemar D-vs-E / D-vs-C + discordant tables	executed	§4	run on grounded typecheck, not the preregistered endpoint
10-min wall clock	executed	600 s timeouts	—
Pre-campaign API requirements 1–8	executed	implemented before campaign	—
30-turn budget	modified	advisory only; overruns C 50/D 38/E 32, max 88	budget is uncontrolled; §4.3 sensitivity analysis
Determinacy pre-screen (exclude)	modified	post-hoc dual annotation (79%, $\kappa \approx 0.20$); 74-task subset, none excluded	subset analysis, not screened population

Preregistered component	Status	What actually happened	Inferential consequence
Arm-D production server	modified	production Worker code served locally over shipped shards (deviation 2)	none material
Single-shot elaboration grading	modified	persistent REPL, same pins (deviation 3)	none material
≤ 4 typecheck calls in-loop	modified	no in-loop typechecking at all (deviation 4)	halluc rate is a pure grounding measure
Skill/ToolSearch leak	modified	sealed before eval; 120 dev runs quarantined (deviation 6)	dev split discarded, eval clean
Per-arm cost distributions faithful@budget (BEq+ equivalence)	modified not executed	means only (USD + wall-clock) grader is a stub; zero judge files in campaign	descriptive primary endpoint missing; no confirmatory claim possible
LLM judge + 50-item human calibration	not executed	harness existed unused; an <i>uncalibrated</i> blind judge pass was launched post-review (§5)	§5 is exploratory only
tokens-to-solve (half the success criterion)	not executed	never computed (tokens recorded per row)	P2 untested as specified
3 reseeds + pass@k curves	not executed	one run per (task, arm) everywhere	no variance estimate; single-seed caveat on every number
Second model class on primary set	not executed	Tier-1 all-Haiku; v2 all-Sonnet on different benchmarks	capability-band generality unknown
Preregistered success criterion	not executed	neither half graded as specified	zero confirmatory results
Tier 2 as specified (FATE-H + MPR-Prop proving, reflection loop, arms A/C/D/E)	not executed	replaced by exploratory v2: QR/MPR retrieval + one-shot SorryDB, arms N/F/W/WF	§§6–7 are exploratory, not the preregistered Tier 2
PutnamBench	not executed	—	—
Tier 3a offline Erdős set	not executed	—	—
Tier 3b live Erdős queue	not executed	—	—

C Data file map

Path (preservation repo)	Contents	Key provenance commits (WikiLean)
docs/research/BRIDGE-EXPERIMENT.md docs/research/BRIDGE-RESULTS.md docs/research/review/	preregistration incl. deviations 1–7 Tier-1 + retrieval results log external review 1 verbatim + independent claim-by-claim verification (29 confirmed / 1 partial / 1 refuted) Tier-1 harness: runner, scorer, REPL typecheck rig, trace analysis, arm manifests	0d36f266 (2026-07-16) 53f5cb31, daac6107 f97e67e4
bench/*.py, bench/README.md, bench/arms/		068abe34, e554902e, 6c2ca704, 31b95caa, 79ac3dfe
bench/data/bridge_tasks.jsonl (+.stats) bench/data/fresh_tasks.jsonl (+.stats)	371 ProofNet# tasks; source pin + MIT licence 100 post-Brain-index tasks + determinacy annotations + the (Brain-scoped) held-out check	e554902e, ec2a9e11 df5dbf92, a0d45103
bench/data/bridge_summary.json bench/data/runs/ bench/data/runs_E_fresh_429_archive/ bench/data/runs_devleak_2026-07-18/ bench/analysis/	v1 Tier-1 scored summary (pre-repair) all 2,355 Tier-1 run rows, incl. the 31 repaired E rows the 31 original 429-errored E rows, byte-preserved the 120 quarantined skill-leak dev runs (deviation 6) v2 corrective statistics: tier1_reanalysis.{py,json,md}, fresh_exposure.{py,json,md}, retrieval_clustered.{py,json,md}, rerun_E_fresh.py, score_e31_v2.py → bridge_summary_v2.json + part1_fresh100_v2.{json,md}, judge_fresh_run.py / judge_fresh_summary.py	53f5cb31 19a90209 (repair driver) f97e67e4 f89e7a41 64c48052, 19a90209
bench/v2/ bench/v2/data/	v2 harness + AGENT_MANUAL.md MathlibQR/MPR (LeanSearch-v2 @94f4888cbaf9, CC BY 4.0), SorryDB-2601 split, tasks_frozen.jsonl	c7629584, 21032c06 c7629584, 87638cfc, db679e8d
bench/v2/runs/	every v2 run row with full gzipped stream-json transcripts (QR 810×4, MPR 69×4, SorryDB 508 rows) + verify.jsonl — all 203 frozen-candidate kernel verdicts, complete	3664aa3d, daac6107, 382e51bf, 1fe21cca, 9682b3b1

Recomputation entry points: bench/score_bridge.py (Tier-1 grading), bench/analysis/tier1_reanalysis.py + score_e31_v2.py (§4), bench/analysis/fresh_exposure.py (§4.4), bench/analysis/retrieval_clustered.py (§6.1–6.2, §7 uncertainty), bench/trace_analysis.py (§4.5), bench/v2/score_retrieval.py (§6 point estimates), bench/v2/verify_sorrydb.py (§7 verdicts), bench/analysis/judge_fresh_summary.py (§5, pending). **External sources cited** (fetched and verified during the v2 design sweep): TheoremGraph arXiv:2606.25363 · LeanSearch arXiv:2403.13310 · LeanSearch-v2 arXiv:2605.13137 + github.com/frenzymath/LeanSearch-v2 · LeanExplore arXiv:2506.11085 · SorryDB arXiv:2603.02668 + sorrydb.org · FATE arXiv:2511.02872 · miniCTX arXiv:2408.03350 · LeanDojo arXiv:2306.15626 · LeanAgent arXiv:2410.06209 · Numina-Lean-Agent arXiv:2601.14027 · miniF2F-v2 arXiv:2511.03108 · ProofNet#/ProofNetVerif arXiv:2406.07222 · benchmark-faults survey arXiv:2606.29493.